CHAPTER 1

THE PROBLEM AND ITS BACKGROUND

Introduction

In the Philippine context, Autism Spectrum Disorder (ASD) is a neurodevelopmental condition characterized by persistent difficulties in social communication, restricted interests, and repetitive patterns of behavior. It is considered a complex developmental condition that affects multiple aspects of a child's growth, including cognitive, emotional, and social functioning (Department of Health, Philippines, 2020). ASD presents with varying levels of severity, where children may exhibit differences in communication abilities, behavioral regulation, and social interaction (Cuaso-Lim & Tan-Ting, 2020). These characteristics make early identification essential, as developmental differences often emerge during early childhood and may influence long-term outcomes if not addressed appropriately.

Autism Spectrum Disorder (ASD) is recognized as a significant developmental concern, with prevalence estimates suggesting that approximately 1.2 million Filipinos may be on the autism spectrum, including both diagnosed and undiagnosed individuals (Reyes & Domingo, 2025; Capitol Medical Center, 2025). However, precise nationwide data on diagnosed Autism Spectrum Disorder cases in the Philippines remains limited due to the absence of a centralized autism registry and large-scale epidemiological monitoring systems. As a result, many prevalence figures are based on estimates derived from international prevalence rates rather than confirmed national case counts.

Early identification is particularly important during early childhood, including the toddler stage, as this period is critical for detecting developmental differences associated with Autism Spectrum Disorder (Huda et al., 2024; Baron-Cohen et al., 2021). During this stage, children begin to exhibit observable patterns in communication, social interaction, and behavior that are essential for autism screening (Robins et al., 2023). Since toddlers cannot independently respond to assessment tools, parents or legal guardians serve as the primary respondents, as their daily interaction with the child allows them to provide reliable observations. Therefore, parent-reported data is considered a valid and necessary source for early screening (Robins et al., 2023).

ASD affects not only individuals but also families and communities. Research involving Filipino families shows that children with ASD often experience challenges in communication, behavior, and daily functioning, requiring continuous support and intervention (Reyes & Domingo, 2025). Similarly, studies on Filipino parents and caregivers reveal emotional, social, and practical challenges, including understanding behavioral patterns and accessing appropriate services (Roxas et al., 2022). These findings emphasize the broader impact of ASD and the need for accessible and responsive support systems.

Despite increasing awareness, several challenges remain in the Philippine context. Many families experience difficulties in recognizing early signs of ASD due to limited knowledge and understanding of developmental conditions (Roxas et al., 2022). Access to specialized healthcare services is

also uneven, particularly in resource-limited areas. Although policies for ASD care and screening exist, issues in implementation and accessibility persist (Department of Health Philippines, 2020). These limitations contribute to delays in diagnosis and intervention.

Another major concern is the financial and logistical burden associated with autism assessment and care. Families often face economic strain due to the costs of medical consultations, therapy sessions, and specialized education (Reyes & Domingo, 2025). As a result, many parents rely on personal observations and informal sources, which may further delay professional consultation and early intervention. These conditions highlight the need for a locally developed, accessible screening system tailored to the Philippine setting.

In response to these challenges, advancements in technology offer new opportunities to improve healthcare accessibility. Machine learning has been widely applied in various studies to analyze behavioral data and identify patterns associated with developmental conditions, improving the efficiency and consistency of early identification (Tartarisco et al., 2021; Sollis et al., 2025). In the Philippine setting, studies such as de Vera et al. (2024) demonstrate the potential of intelligent systems in monitoring and predicting autism-related behaviors. At the same time, Natural Language Processing (NLP) preprocessing techniques such as text normalization, keyword extraction, and behavioral mapping can support guided conversational systems by improving the processing and interpretation of parent-provided observations.

The integration of machine learning and NLP presents a promising approach to autism screening, particularly in resource-limited settings. Guided conversational systems can assist parents in describing their child's behavior, navigating structured screening questionnaires, and receiving preliminary behavioral insights through accessible and user-friendly interaction.

These tools have the potential to reduce barriers related to accessibility, cost, and awareness, especially in communities where formal screening services are not readily available.

However, despite the availability of screening tools and emerging technologies, there remains a lack of localized, accessible, and user-centered pre-screening systems in the Philippines that integrate intelligent data analysis with guided user interaction. Existing tools are often not easily accessible or require professional administration, limiting their use for early detection.

Therefore, this study proposes the development of **AuRA (Autism Risk Analytics)**, a machine learning-assisted Autism Spectrum Disorder (ASD) pre-screening and educational support system for toddlers. The proposed system utilizes NLP preprocessing, rule-based behavioral indicator extraction, ordinal feature encoding, and machine learning-based classification to assist in identifying possible ASD-related behavioral indicators from parent-provided observations through a structured and guided assessment interface.

This study aims to develop a system that supports early awareness and screening by combining validated behavioral indicators with intelligent

data analysis and user-friendly interaction. Ultimately, the study seeks to contribute to improving early detection practices and supporting families in making timely decisions regarding professional consultation and intervention.

Background of the Study

Autism Spectrum Disorder (ASD) remains a significant concern in the Philippines, particularly in terms of early detection and intervention. Local studies indicate that many families face challenges not only in caring for children with ASD but also in accessing appropriate and timely healthcare services. According to Reyes and Domingo (2025), Filipino families raising children with ASD commonly experience financial constraints, limited access to healthcare, and delays in obtaining a formal diagnosis. Despite the increasing level of awareness, these challenges continue to hinder effective early detection and intervention.

In addition, existing studies highlight the difficulty experienced by families in recognizing early signs of ASD. Santos and Zulueta (2024) reported that family members often have limited understanding of ASD-related behaviors in young children. Similarly, Roxas et al. (2022) found that parents frequently struggle to determine whether certain behaviors are part of typical development or indicators of autism. Furthermore, Makino et al. (2021) emphasized the critical role of parents in early detection, as their daily interaction with the child enables them to provide valuable observations for screening purposes.

Although numerous studies on ASD have been conducted in the Philippine context, most of these focus on diagnosis, causes, and family

experiences. Studies such as those by Dy et al. (2023) and Cuaso-Lim and Tan-Ting (2020) do not primarily address the development of simple and accessible screening tools that parents can use for early identification. This indicates a gap between existing research and its practical application in real-life settings.

Moreover, most ASD screening tools are administered by healthcare professionals. While these tools are generally reliable, they are not always accessible due to factors such as cost, availability, and limited public awareness. As a result, many children are only assessed when developmental delays have already become evident, which may negatively affect early intervention outcomes. Additionally, there is a lack of user-friendly tools that parents can utilize at home for preliminary screening.

Recent advancements in technology, particularly in machine learning and Natural Language Processing (NLP), offer potential solutions for improving early ASD detection. Studies conducted in other countries demonstrate that machine learning can effectively identify autism risk based on behavioral data. However, in the Philippine context, the application of these technologies remains limited, especially in developing tools designed for parent use. Existing systems often prioritize accuracy but do not sufficiently address usability and accessibility.

Given these challenges, there is a need for a system that integrates clinical knowledge with user-friendly design. A system that combines established ASD indicators, machine learning techniques, and an accessible interface may assist parents in conducting early screening. In addition, a

chatbot-based system can provide guided interaction, enabling parents to respond to screening questions in a structured and understandable manner.

Therefore, this study aims to develop a system that assists parents in identifying possible signs of autism in toddlers. The proposed system emphasizes ease of use, accessibility, and practical support for early guidance. It is intended solely as a pre-screening tool and not as a substitute for professional diagnosis. Ultimately, this study seeks to contribute to improved early detection and encourage timely consultation with healthcare professionals.

Statement of the Problem

This study aims to address the challenges in the early identification of Autism Spectrum Disorder (ASD) in toddlers (12 to 36 months), particularly in communication, social interaction, and behavioral development, through the development of a machine learning-based pre-screening system that supports early detection of possible ASD signs.

Specifically, this study seeks to answer the following questions:

1. How can a structured screening instrument, adapted from validated tools such as M-CHAT-R/F, Q-CHAT, and CSBS-DP Infant-Toddler Checklist, be developed to effectively collect data on early autism-related behaviors in toddlers as reported by parents?
2. How can Random Forest, Logistic Regression, and XGBoost be integrated using a Soft Voting Ensemble to classify toddler behavioral data into low, moderate, and high ASD risk levels, and how does their

performance compare in terms of accuracy, precision, recall, and F1-score?

3. How usable and effective is the AuRA system for parents and non-expert users in conducting guided ASD pre-screening through structured interaction and machine learning-assisted risk classification?

Objectives of the Study

General Objective

The general objective of this study is to design and evaluate AuRA, a machine learning-assisted Autism Spectrum Disorder (ASD) pre-screening system that utilizes NLP preprocessing techniques, behavioral feature extraction, and guided interaction to assist in identifying possible ASD-related behavioral indicators in toddlers aged 12–36 months.

Specific Objectives

1. To design a structured screening instrument and dataset by adapting items from validated tools such as M-CHAT-R/F, Q-CHAT, and the CSBS-DP Infant-Toddler Checklist in order to effectively collect parent-reported data on early autism-related behaviors in toddlers.
2. To utilize and evaluate Random Forest, Logistic Regression, and XGBoost, and integrate their outputs using a soft voting ensemble for ASD risk classification, while determining their performance based on accuracy, precision, recall, and F1-score.
3. To assess the usability and real-world effectiveness of the AuRA system for parents and non-expert users in conducting guided ASD

pre-screening through structured interaction, machine learning-assisted risk classification, and result visualization.

Theoretical Framework

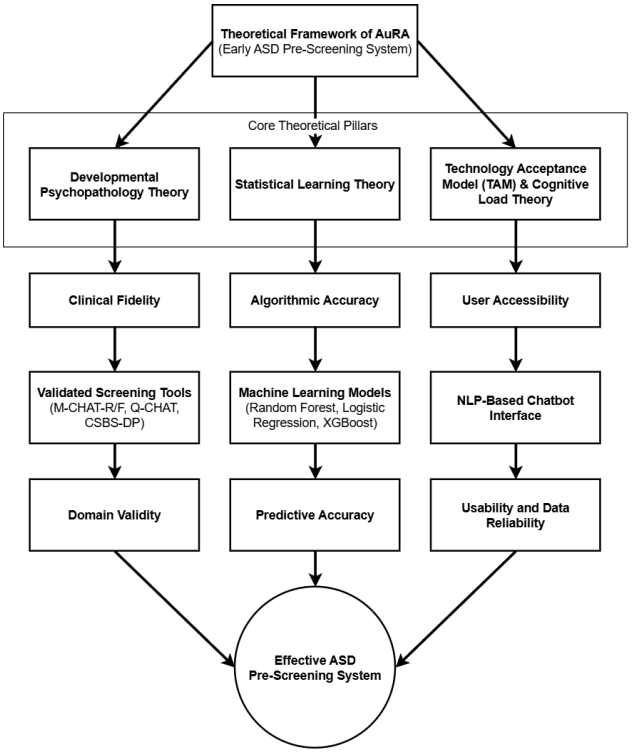


Figure 1.1 Theoretical Framework Diagram

The proposed system, AuRA, is grounded in a multidisciplinary framework that integrates developmental psychology, machine learning, and human-computer interaction to support early Autism Spectrum Disorder (ASD) screening in a practical and accessible manner.

The study is first anchored in Developmental Psychopathology Theory, which explains that ASD is a developmental condition characterized by deviations from typical patterns of growth, particularly in communication, social interaction, and behavior (Huda et al., 2024). Early childhood,

especially between 12 to 36 months, is considered a critical period for identifying these deviations due to rapid cognitive and social development (Baron-Cohen et al., 2021; Robins et al., 2023). This supports the focus of the study on early screening, as early identification allows timely intervention and improved developmental outcomes.

To ensure clinical relevance, the system utilizes a combined screening instrument adapted from validated tools such as M-CHAT-R/F, Q-CHAT, and CSBS-DP Infant-Toddler Checklist, along with additional questionnaire items. The questionnaire was further refined through expert consultation and pilot testing to assess the clarity and understandability of the questions among parent respondents. Feedback from the pilot testing was used to clarify wording, add concrete examples, and improve the structure of the questions and follow-up responses. These tools provide established behavioral indicators associated with ASD and serve as the basis for constructing the dataset used in the system. The inclusion of multiple screening instruments expands the scope of behavioral variables and improves the comprehensiveness of data collection (Sobieski et al., 2024; Centers for Disease Control and Prevention, 2025). The questionnaire was reviewed with the assistance of a healthcare professional to assess the relevance and clarity of the behavioral indicators and questionnaire items.

The study is also guided by modern applications of statistical learning principles, which explain how machine learning models can learn patterns from data to make predictions. Since behavioral data in ASD screening may be complex and non-linear, the system applies Logistic Regression, Random

Forest, and XGBoost (Saihi et al., 2021; Rubio-Martín et al., 2024). These supervised learning algorithms are used to classify behavioral inputs into Low, Moderate, and High ASD risk levels. Their predicted class probabilities are combined through a soft voting ensemble strategy to improve the robustness and stability of the final classification. The performance of the individual models and the ensemble classifier are evaluated using metrics such as accuracy, precision, recall, and F1-score.

In addition, the study considers the presence of response bias, particularly in parent-reported data. Caregivers may experience emotional stress, uncertainty, or external influences that affect how they perceive and report a child's behavior (Reyes & Domingo, 2025). This is addressed through structured questionnaire design and guided interaction to improve the consistency and reliability of responses.

To enhance usability, the system incorporates principles aligned with recent human-computer interaction and user acceptance studies, emphasizing that systems are more likely to be adopted when they are easy to use and useful (Deng et al., 2024). Additionally, reducing cognitive load improves user understanding and interaction efficiency in digital systems (Huda et al., 2024). Based on these principles, the system integrates a guided conversational interface supported by NLP preprocessing techniques such as text normalization, rule-based behavioral indicator extraction, and behavioral mapping. These techniques help improve the consistency and interpretability of parent-provided observations before machine learning classification.

Finally, the system aligns with modern Clinical Decision Support System (CDSS) approaches, which focus on providing structured and evidence-based recommendations to assist decision-making in healthcare (Centers for Disease Control and Prevention, 2025). The system translates model outputs into categorized risk levels and provides guidance for further assessment or consultation, supporting early decision-making for parents.

Conceptual Framework

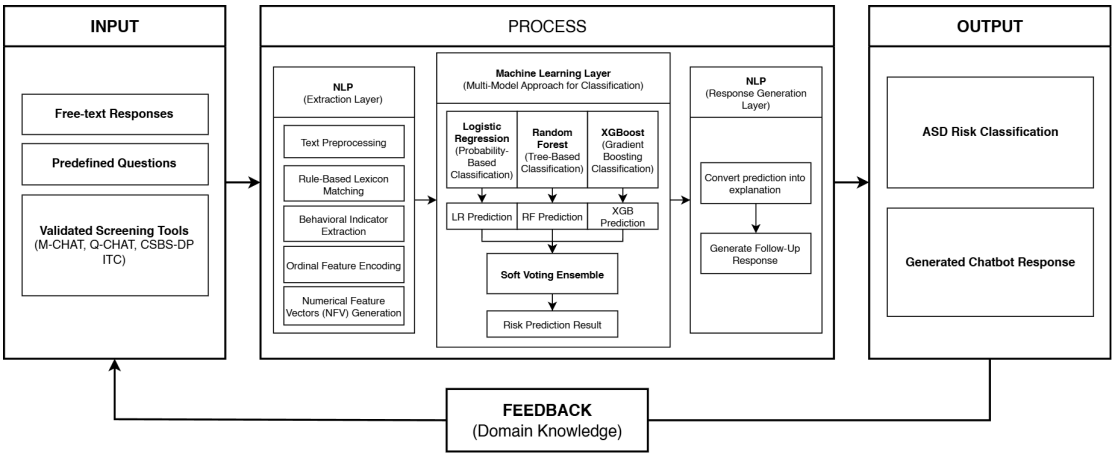


Figure 1.2 Conceptual Framework Diagram

Figure 1.2 illustrates the conceptual framework of the AuRA (Autism Risk Analytics) system, which is structured according to the Input-Process-Output (IPO) model with an integrated feedback mechanism. This framework depicts the systematic flow of information, from the initial collection of user data to the final generation of a classification result and its corresponding conversational response, while ensuring that the entire process remains anchored to established clinical and domain-specific knowledge.

The Input stage collects parent-reported behavioral observations through the structured questionnaire, consisting of responses to the main questions and corresponding open-ended follow-up questions. The questionnaire incorporates behavioral indicators informed by M-CHAT-R/F, Q-CHAT, and CSBS-DP Infant-Toddler Checklist. The inclusion of these three input modalities establishes the clinical validity and comprehensiveness of the data being collected, ensuring that the information gathered is relevant, structured, and methodologically consistent with established ASD screening practices used in developmental pediatrics.

Once collected, the input data undergo a three-layered process consisting of interconnected computational components. The first component, the Natural Language Processing (NLP) Extraction Layer, is responsible for transforming unstructured or semi-structured input into a machine-readable format. This layer performs text preprocessing (e.g., tokenization, normalization, and noise removal), rule-based lexicon matching to identify clinically relevant keywords and phrases associated with behavioral indicators of ASD, and the extraction of these behavioral indicators from the parent or caregiver's responses. The extracted behavioral indicators are assigned ordinal numerical values through Ordinal Feature Encoding and converted into Numerical Feature Vectors (NFV) for machine learning classification.

The second component, the Machine Learning Layer, applies a multi-model ensemble approach to classification, which improves classification robustness by combining the strengths of multiple supervised learning algorithms. In this layer, three independent classification

models—Logistic Regression, Random Forest, and XGBoost—process the same numerical feature vector input in parallel rather than sequentially. Logistic Regression generates probability-based estimates of ASD risk; Random Forest, a tree-based ensemble algorithm, captures non-linear relationships among behavioral indicators; and XGBoost (Extreme Gradient Boosting) iteratively improves prediction accuracy through gradient boosting techniques. Each model independently produces its own predicted class probabilities (i.e., LR, RF, and XGB predictions) without relying on the outcome of the other models. These three outputs are subsequently aggregated using a soft voting ensemble method, wherein the final classification is determined by the highest average probability across the three models, a technique that improves the robustness of the system's final classification compared with relying on a single classifier.

The third component, the NLP Response Generation Layer, receives the final classification output from the ensemble model and maps it to a predefined, clinically reviewed explanation and follow-up message template. This layer ensures that the system's output is communicated to the end user (e.g., a parent or caregiver) in a manner that is both clinically accurate and conversationally accessible, using structured and pre-validated response content rather than open-ended or free-form text generation.

The Process stage culminates in the Output stage, which yields two complementary outputs: the ASD Risk Classification, representing the system's predicted ASD pre-screening risk level, and the Generated Chatbot Response, which presents this determination to the user in natural,

comprehensible language accompanied by relevant explanatory context. Together, these outputs ensure that the system delivers not merely a numerical or categorical result, but a meaningful and interpretable outcome that supports informed decision-making on the part of the user, while also serving an educational role by helping parents or caregivers better understand ASD-related behavioral indicators and the significance of the screening result.

Completing the framework is the Feedback component, which is composed of two distinct but complementary knowledge bases: the ASD Domain Knowledge base and the Medical Library. The ASD Domain Knowledge base serves as the clinical foundation for the system's computational components, providing the behavioral indicators, lexicons, and screening criteria used by the NLP Extraction Layer and the Machine Learning Layer to identify and classify ASD-related patterns in parent or caregiver responses. This ensures that the system's risk assessment functions remain grounded in current and clinically validated behavioral frameworks. The Medical Library, by contrast, functions as the informational reference used by the chatbot to respond to ASD-related inquiries raised by the user, supplying evidence-based information curated from peer-reviewed literature and established clinical references.

By maintaining these two knowledge bases as separate components, the system achieves a modular architecture in which updates to one do not inadvertently affect the other. For instance, the addition of new general ASD information to the Medical Library does not alter or interfere with the behavioral indicator matching performed by the ASD Domain Knowledge base

within the Machine Learning Layer. This separation not only improves the maintainability of the system but also reinforces the distinct clinical and functional purposes of the risk classification process and the chatbot's educational role, respectively.

Significance of the Study

This study is important because it provides a technology-based approach to help parents and professionals identify possible early signs of Autism Spectrum Disorder. It serves as a bridge between parents' initial concerns and professional consultation through a simple and easy-to-use system.

Parents or Guardians – The system offers an accessible, low-cost, and private tool that allows parents to observe and assess their child's behavior at home. It can help reduce uncertainty by providing immediate feedback and guidance on whether further consultation may be needed.

Healthcare Practitioners – The system may serve as an initial screening support tool that provides organized behavioral information and possible risk levels. This can help professionals prepare for a more detailed assessment.

Educational Institutions – Teachers and school administrators may use the system to better understand students who may show early signs of autism and provide appropriate support.

Information Technology Field – This study contributes to the development of digital healthcare support systems by demonstrating how machine learning

models, behavioral feature extraction, and NLP preprocessing techniques can be integrated into a structured and user-friendly pre-screening system.

Future Researchers – This study may serve as a reference for future research on pre-screening tools and the use of machine learning in developmental conditions

Scope and Limitations of the Study

This study focuses on the development of AuRA, a machine learning-based pre-screening system for Autism Spectrum Disorder (ASD) in toddlers. The system is designed for toddlers aged 12 to 36 months, with parents serving as the primary users who provide behavioral observations of their children. It integrates a guided conversational interface supported by NLP preprocessing techniques to assist users in completing structured screening questions. M-CHAT-R/F, Q-CHAT, and the CSBS-DP Infant-Toddler Checklist are adapted for the questionnaires. The system applies three machine learning algorithms—Random Forest, Logistic Regression, and XGBoost—whose predictions are combined using a Soft Voting Ensemble to produce the final classification of behavioral data into three ASD risk levels: Low, Moderate, and High. In addition, the system presents screening results through visualizations, such as charts, to help users better understand the outcomes. The chatbot also provides educational information about Autism Spectrum Disorder (ASD) using curated medical references to help parents or caregivers better understand the screening outcome and ASD-related concepts.

However, this study has several limitations. The system functions only as a pre-screening tool and does not provide a medical diagnosis. It is not intended to replace professional assessment conducted by licensed healthcare professionals. Furthermore, the educational information provided by the chatbot is limited to curated ASD-related references included in the system's Medical Library and should not be considered a substitute for professional medical advice or consultation. The accuracy of the results depends on the honesty and correctness of the responses provided by parents or legal guardians. The performance of the machine learning models is also limited by the quality and size of the dataset used for training and validation. Furthermore, the system requires internet connectivity to support online system access and guided interaction features.

Although the general target population is toddlers aged 12 to 36 months, the findings of the study may still be limited by the sample size, respondent participation, and the quality of the collected behavioral data. Lastly, the study is limited to the use of Random Forest, Logistic Regression, and XGBoost integrated through a Soft Voting Ensemble, and does not evaluate other machine learning algorithms or ensemble strategies.

Definition of Terms

For clarity, the following terms are defined as used in this study:

Accuracy – A performance metric that measures the proportion of correct predictions (both true positives and true negatives) made by a machine learning model out of all predictions.

AuRA (Autism Risk Analytics) – A chatbot-based pre-screening system that utilizes machine learning and Natural Language Processing (NLP) to assist in identifying possible signs of Autism Spectrum Disorder (ASD) in toddlers through parent-reported behavioral data.

Autism Spectrum Disorder (ASD) – A developmental condition that affects social interaction, communication, and behavior.

Behavioral Indicator – An observable action, response, or pattern in a child's communication, social interaction, or behavior that may signal a developmental concern such as ASD.

Chatbot – A software application that can simulate human conversation through text. In this study, it helps users complete the questionnaire and answers questions about autism.

Clinical Decision Support System (CDSS) – A system designed to provide structured, evidence-based information or recommendations that assist healthcare-related decision-making.

CSBS-DP Infant-Toddler Checklist – A validated screening tool used to assess communication and symbolic behavior in infants and toddlers, used as one of the bases for AuRA's questionnaire items.

F1-Score – A performance metric that combines precision and recall into a single value, used to evaluate the balance between the two when assessing a model's classification performance.

Logistic Regression – A statistical and machine learning method used to classify data into two or more categories by estimating probabilities.

M-CHAT-R/F – The Modified Checklist for Autism in Toddlers, Revised with Follow-Up; a widely used validated screening tool for identifying early signs of ASD in toddlers.

Machine Learning (ML) – A field of artificial intelligence that creates systems capable of learning from data and improving performance over time without being explicitly programmed.

Natural Language Processing (NLP) – Computational techniques used in AuRA to preprocess parent-provided text responses and extract relevant behavioral indicators through rule-based text processing.

Numerical Feature Vector (NFV) – A structured, quantifiable representation of extracted behavioral indicators used as input for machine learning classification.

Ordinal Feature Encoding – A method of representing identified behavioral indicators using ordered numerical values, where feature values of 0, 1, and 2 correspond to increasing levels of the defined behavioral observation or screening concern.

Precision – A performance metric that measures the proportion of correctly identified positive cases out of all cases predicted as positive by a model.

Pre-screening – A preliminary, non-diagnostic assessment used to identify individuals who may show early signs of a condition and who may benefit from further professional evaluation.

Q-CHAT – The Quantitative Checklist for Autism in Toddlers; a validated screening tool used to assess autism-related traits in young children.

Random Forest – A machine learning technique that combines many decision trees to make more accurate predictions by choosing the most common output among them.

Recall – A performance metric that measures the proportion of actual positive cases that were correctly identified by a model.

Risk Level (Low, Moderate, High) – A categorical classification used by the AuRA system to indicate the likelihood of ASD-related behavioral concerns based on parent-reported data.

Screening – The process of identifying individuals who may be at risk for a condition, used in this study as a preliminary step before formal diagnosis.

Soft Voting Ensemble -- A machine learning ensemble approach that combines the predicted class probabilities of multiple models by averaging them, with the final classification determined by the class that has the highest average probability.

Text Normalization – An NLP preprocessing technique that converts text into a consistent, standard format (e.g., lowercasing, removing punctuation) to improve accuracy in keyword extraction.

Toddler – In this study, a child between 12 to 36 months of age, representing the target population for early ASD screening.

Usability – The extent to which a system is easy to learn, understand, and use effectively by its intended users, such as parents or non-expert users.

XGBoost (Extreme Gradient Boosting) – A highly efficient machine learning method used for classification that combines many models to achieve accurate predictions. An ensemble technique that builds models sequentially to correct errors of previous models.

CHAPTER 2

REVIEW OF RELATED LITERATURE

This chapter presents the review of related literature and studies about Autism Spectrum Disorder (ASD), with focus on early detection among toddlers. It includes both foreign and local sources that discuss screening methods such as parent questionnaires and standard tools, as well as the use of modern technologies like machine learning, natural language processing, and chatbot systems in autism detection.

The reviewed literature shows the importance of early identification, the effectiveness of screening tools, and the growing use of digital technologies to improve the accuracy and accessibility of ASD assessment. It also presents the current situation in the Philippines, including challenges in access to services and early screening. These studies serve as the foundation of the present research.

Table 2.1: Foreign Literature

	Title/Authors	**Methods**	**Conclusion**	**Objectives**	Recommendations
1	**Romero-García et al. (2021)**	NAO robot + Q-CHAT interaction	Robotics increases engagement	Explore robotic screening	Integrate robotics in ASD screening
2	**Deng et al. (2024)**	AI chatbot for autism interaction	Chatbot supports ASD detection	Develop conversational AI system	Use AI in early screening

	Title/Authors	Methods	Conclusion	Objectives	Recommendations
3	**Saihi et al. (2021)**	Machine learning classification	ML improves ASD prediction accuracy	Build prediction model	Combine ML + questionnaires
4	**Rubio-Martín et al. (2024)**	NLP-based autism detection system	NLP improves early detection	Apply NLP in screening	Integrate AI in healthcare
5	**Centers for Disease Control and Prevention (2025)**	Review of standardized screening tools used in early ASD detection	CSBS is a recognized tool for assessing early communication and social behaviors	To provide guidelines for early screening and identification of ASD	Use validated tools such as CSBS for early developmental screening
6	**Huda et al. (2024)**	Systematic review of 51 studies using multiple databases analyzing validity, reliability, and accuracy of ASD screening tools across cultures	Screening tools vary in effectiveness across cultures; tools perform best when culturally adapted	To evaluate ASD screening tools in culturally diverse populations	Develop culturally sensitive and population-specific screening tools
7	**Zhang (2025)**	Review of AI + ML methods:	AI and ML significantly improve efficiency of	Summarize AI-assisted methods for early	Build large shared

	Title/Authors	Methods	Conclusion	Objectives	Recommendations
		predictive behavioral models, brain imaging analysis, intelligent screening systems for infants, toddlers, preschoolers	early screening and intervention; traditional methods cause delays in diagnosis past age 6	screening, diagnosis, and intervention of autism specifically in young children	datasets; develop AI tools tailored to toddler age groups for mass early screening
8	**Sideraki & Anagnostopoulos, Mujeeb Rahman & Monica Subashini, Lu et al (2025)**	Deep learning, SVM, Random Forest, CNN, RNN; biometric data, video-based interaction, linguistic feature extraction; ML chatbots; AAC systems	AI automates diagnostic tools, reduces subjective bias; ML chatbots provide personalized language support; deep learning identifies behavioral patterns more accurately than traditional methods	Determine how AI can be leveraged for accurate ASD assessment, personalized intervention, and long-term monitoring	Further research on long-term evaluation and customization; integrate chatbots and AAC tools into clinical ASD workflows
9	**Hasan (2025)**	Systematic review of AI-driven tools	AI tools achieve diagnostic accuracies exceeding	Assess how AI tools detect subtle ASD markers	Address data privacy and bias issues; design

	Title/Authors	Methods	Conclusion	Objectives	Recommendations
		(ML, deep learning) detecting gaze anomalies, motor irregularities, neurophysiological patterns in children from birth to preschool age	84%; integrating AI into clinical workflows still challenged by data privacy, model bias, and scalability	during the birth-to-preschool window where interventions have greatest impact	scalable AI tools for early ASD detection in low-resource clinical settings
10	**Chaxiong, Dimian, & Wolff (2022)**	Systematic review of 41 studies covering children from birth through age three, examining restricted and repetitive behavior (RRB) patterns	Differences in the frequency and intensity of repetitive behaviors are observable before age two; findings on sensory behaviors and insistence on sameness remain mixed	To synthesize evidence on restricted and repetitive behavior in children with autism during the first three years of life	Use RRB indicators as supporting early markers, but avoid treating the quantity of repetitive behaviors as a direct measure of autism severity
11	**Wieckowski et al. (2023)**	Systematic review and meta-analysis of 50 studies on the	Pooled sensitivity was 0.83 and specificity 0.94; performance varied	To determine the diagnostic accuracy of the M-CHAT (original	Retain structured follow-up questions in screening tools; interpret reported

	Title/Authors	Methods	Conclusion	Objectives	Recommendations
		sensitivity and specificity of the M-CHAT and M-CHAT-R/F, including children screened up to 48 months	depending on sample and screening procedures, with the follow-up component proving essential	and revised) as a toddler ASD screening instrument	accuracy as specific to the instruments reviewed

Table 2.2: Local Literature

	Title/Authors	Methods	Conclusion	Objectives	Recommendations
1	**Balbin et al. (2022)**	Literature review with content analysis	Adults with autism face challenges in social integration,healthcare access, and overall quality of life	To analyze factors affecting the quality of life of adults with autism	Improve support systems, accessibility to services, and policy implementation.
2	**Chiu et al. (2023)**	Q-methodology mixed-method approach	Barriers include socioeconomic constraints, systemic discrimination, and lack of trust in healthcare services.	To examine barriers faced by Filipino families in accessing rehabilitation services for children with special needs.	Improve accessibility, integrate telehealth, and develop inclusive and culturally responsive healthcare programs.

	Title/Authors	Methods	Conclusion	Objectives	Recommendations
3	**Santos & Zulueta (2024)**	Qualitative phenomenological study using in-depth interviews of individuals with ASD siblings	Individuals experience emotional stress, responsibility, and social challenges but also develop empathy and resilience	To explore lived experiences of individuals with ASD siblings	Strengthen family support systems and awareness programs

Table 2.3: Foreign Studies

	Title/Authors	Methods	Conclusion	Objectives	Recommendations
1	**Baron-Cohen et al. (2021)**	Large-scale toddler screening using Q-CHAT questionnaire	Q-CHAT is effective for early ASD risk detection	To validate Q-CHAT in population screening	Use repeated screening across early development stages
2	**Allison et al. (2020)**	Questionnaire-based screening for toddlers aged 18–24 months.	Early scoring detects ASD risk effectively	To evaluate early ASD detection capability	Implement Q-CHAT in pediatric screening programs.
3	**Rutter et al. (2022)**	Comparison of two ASD screening tools	Both tools effective but Differ in sensitivity.	To compare screening performance	Combine tools for improved detection accuracy.

	Title/Authors	Methods	Conclusion	Objectives	Recommendations
4	**Robins et al. (2023)**	Parent-reported screening questionnaire validation.	M-CHAT-R/F is reliable across cultures.	To validate screening tool usability.	Adapt screening tools to population context.
5	**Sobieski et al. (2024)**	Population-based study with 602 children using CSBS-DP ITC and statistical analysis for screening accuracy.	CSBS-DP ITC is effective for early ASD screening with good accuracy.	To evaluate the validity and screening accuracy of CSBS for ASD detection.	Use CSBS as an early screening tool; further refinement is needed.
6	**Sollis, Wall, & Washington (2025)**	Machine learning models (Decision Tree, Random Forest, XGBoost) applied to QCHAT-10 dataset; feature selection (RFE); cross-validation using datasets from New Zealand, Saudi Arabia,	ML models achieved up to 91% AUC, showing accurate classification of ASD risk using fewer screening items.	To develop an efficient ML-based ASD screening system using reduced questionnaire items.	Use ML-based screening tools to improve early detection and reduce screening complexity.

	Title/Authors	Methods	Conclusion	Objectives	Recommendations
		and Poland.			
7	**Tartarisco et al. (2021)**	5 ML algorithms: SVM, Random Forest (RF), Naïve Bayes (NB), Logistic Regression (LR), K-Nearest Neighbors (KNN) applied to Q-CHAT toddler dataset	SVM achieved 95% accuracy (best); overall ML system reached 90%; Q-CHAT confirmed cross-culturally valid; SVM-RFE reduced to 14 items maintaining 91% accuracy	Investigate accuracy and reliability of the Q-CHAT using ML to classify young autistic children vs. typically developing toddlers	Apply ML to create shorter, faster Q-CHAT versions; deploy as quick, high-performance primary care screening tool for toddlers
8	**Mujeeb Rahman, K.K. & Monica Subashini, M. (2022)**	Two 5-layer Deep Neural Networks (DNN) applied to QCHAT and QCHAT-10 toddler datasets; compared to SVM and traditional ML	QCHAT-10 DNN achieved 98% accuracy, outperforming SVM and standard DNN on full Q-CHAT; deep learning superior to traditional ML for toddler ASD screening	Apply DNN algorithms to automate ASD identification from QCHAT datasets and compare model performance	Prioritize deep neural network approaches over traditional ML for QCHAT-based toddler screening; reduce screening items for efficiency

	Title/Authors	Methods	Conclusion	Objectives	Recommendations
9	**Lu, H. et al. (2024)**	7 supervised ML models applied to CHAT-23 questionnaire; feature engineering reduced to 9 features; children under 36 months	Best model: sensitivity 0.909, specificity 0.922 with 9 features; children under 36 months show higher diagnostic accuracy than older children	Build and evaluate ML models using CHAT-23 for ASD early screening in toddlers; identify most relevant questionnaire features	Target toddler age window for AI screening; refine screening questionnaires using ML feature selection for clinical efficiency
10	**Abbas et al (2020)**	3-module AI system: mobile app parent questionnaire + home video ML analysis + clinician questionnaire; n=375 children aged 18–72 months; multi-site blinded study	Outperformed baseline screeners by 0.35 AUC; for children under 48 months, outperformed best existing screener by 0.18 AUC; demonstrated toddler-specific AI value	Present a multi-modular ML assessment achieving diagnostic-grade reliability for young children 18–72 months	Combine parent-report, video-based, and clinician AI modules for high-accuracy toddler ASD assessment; deploy via mobile app

	Title/Authors	Methods	Conclusion	Objectives	Recommendations
11	**Chang et al. (2021)**	Computational gaze analysis of 993 toddlers aged 17.1–36.9 months while viewing short videos	Toddlers with ASD showed reduced attention toward social stimuli and differences in the coordination of gaze with speech sounds	To measure patterns of gaze in toddlers with ASD using computational methods	Use objective gaze measures to supplement parent report; note that video-based gaze is not equivalent to parent-reported eye contact
12	**Delehanty & Wetherby (2021)**	Home observation of 211 children approximately 20 months old during everyday activities, coding communicative gestures	The ASD group used a smaller variety of gestures and showed lower rates of deictic gestures such as pointing, and fewer vocalizations	To determine the rate of communicative gestures and its relation to developmental outcomes in toddlers with and without ASD	Include gesture variety and purpose in early screening; examine the context and function of communication, not frequency alone

	Title/Authors	Methods	Conclusion	Objectives	Recommendations
13	**Wu et al. (2024)**	Parent-reported gesture inventories of 467 children measured at 12, 18, and 24 months	Children later diagnosed with ASD showed slower growth in deictic and symbolic gestures from 12 to 24 months	To compare early gesture development in infants who later develop autism against those who do not	Monitor gesture growth over time in young toddlers; note that familial-likelihood samples may not generalize to the general population
14	**Miller et al. (2021)**	Longitudinal observational research from infancy through 36 months, including coded object exploration during the toddler period	More frequent unusual visual inspection of objects was observed in children later diagnosed with ASD, predicting diagnosis as early as 9 months	To determine whether repetitive behavior with objects predicts later ASD diagnosis and social behavior	Consider unusual object exploration as an early indicator; avoid interpreting all repetitive play as indicative of ASD

	Title/Authors	Methods	Conclusion	Objectives	Recommendations
15	**Lee & Schertz (2022)**	Analysis of video-recorded caregiver–child interactions of 20 toddlers with autism averaging approximately 25 months of age	Social turn-taking was positively associated with joint attention, while instrumental turn-taking was not	To examine the association of turn-taking functions with joint attention in toddlers with autism	Consider the social purpose of a child's communication in early screening; note the findings do not establish causation or diagnostic accuracy
16	**Zheng et al. (2024)**	M-CHAT-R/F screening of 5,336 children aged 17–20 months across seven primary-care clinics in Singapore	113 children screened positive and 54 met autism diagnostic criteria; those diagnosed accessed intervention at an average age of 22 months; clinicians rated the process as practical	To evaluate the validity and feasibility of using M-CHAT-R/F in routine primary-care clinics in an Asian setting	Integrate structured ASD screening and follow-up into routine toddler healthcare visits

	Title/Authors	Methods	Conclusion	Objectives	Recommendations
17	**Attar et al. (2023)**	Nonrandomized controlled trial comparing electronic M-CHAT-R/F administration with paper questionnaire and telephone follow-up; 17,900 toddlers screened	Electronic administration improved completion of screening and removed the delay between questionnaire and follow-up, though skilled clinical interviews remained valuable for clarifying responses	To validate the electronic M-CHAT-R/F against traditional administration during routine healthcare visits	Adopt digital screening to improve implementation while retaining clinical interpretation and appropriate evaluation

Table 2.4: Local Studies

	Title/Authors	Methods	Conclusion	Objectives	Recommendations
1	**Dy et al. (2023)**	Clinical screening and genetic testing of diagnosed ASD patients.	Screening plays a crucial role in identifying underlying conditions associated with ASD	To screen Filipino children with ASD for Fragile X Syndrome and examine associated clinical characteristics.	Improve early screening and diagnostic processes for developmental disorders.
2			Families	To explore	Improve

	Title/Authors	Methods	Conclusion	Objectives	Recommendations
	Reyes & Domingo (2025)	Phenomenological qualitative analysis.	experience financial burden, emotional stress, limited access to services,and social stigma.	the lived experiences of Filipino families raising children with autism.	access to support services, strengthen intervention programs, and enhance awareness and Support systems.
3	**Quilendrino et al. (2022)**	Cross-sectional survey of 158 parents of children with autism aged 2–6 years, examining expenses for diagnosis, consultations, therapy, education, and medication	Average diagnostic evaluation cost was approximately ₱7,411.80 and average first-year post-diagnosis intervention expense was ₱38,868; costs varied by family income and access to services	To determine the direct cost of autism and its economic impact on the Filipino family	Develop lower-cost and more accessible screening pathways; treat reported figures as descriptive of the study sample, not current prices
4	**Leosala (2023)**	Phenomenological study using semi-structured interviews with 10 Filipino parents of children with ASD	Themes included children's individual needs, access to supportive services, service barriers, information-seeking,	To explore the early intervention experiences of Filipino parents of children with ASD	Strengthen information dissemination and early-intervention access for Filipino families

	Title/Authors	Methods	Conclusion	Objectives	Recommendations
		aged 3–8 years receiving early childhood intervention	parental involvement, and family adjustment		

Synthesis

The reviewed foreign and local literature and studies collectively emphasize the critical importance of early detection of Autism Spectrum Disorder (ASD), particularly during the toddler stage, where timely intervention can significantly improve developmental outcomes. Across the literature, a consistent finding is that while validated screening tools and advanced detection technologies already exist, their accessibility, integration, and practical application remain limited, especially in developing countries such as the Philippines.

International guidelines, such as those from the Centers for Disease Control and Prevention (2025), highlight the importance of standardized screening tools in early ASD identification. Supporting this, several validated instruments have been widely used in research and practice. Baron-Cohen et al. (2021) demonstrated the effectiveness of the Quantitative Checklist for Autism in Toddlers (Q-CHAT), while Robins et al. (2023) confirmed the reliability of the Modified Checklist for Autism in Toddlers, Revised with Follow-Up (M-CHAT-R/F) across cultural settings. Rutter et al. (2022) further emphasized that Q-CHAT-10 and M-CHAT are both effective tools but differ in

sensitivity, suggesting the value of complementary use. In addition, Sobieski et al. (2024) validated the Communication and Symbolic Behavior Scales Developmental Profile Infant-Toddler Checklist (CSBS-DP ITC) as effective in identifying early communication and social behavior deficits.

The accuracy of these instruments has likewise been examined at scale. Wieckowski et al. (2023), in a meta-analysis of fifty studies, reported a pooled sensitivity of 0.83 and specificity of 0.94 for the M-CHAT and its revised version, while noting that performance varied according to sample characteristics and screening procedures, and that the structured follow-up component was essential to accurate results. This finding directly informs the present study's decision to include follow-up questions that appear conditionally based on the parent's initial response. The feasibility of such screening in real service settings has also been established. Zheng et al. (2024) screened 5,336 toddlers aged 17 to 20 months across seven primary-care clinics in Singapore and found that children who screened positive and were later diagnosed accessed intervention at an average age of 22 months, with clinicians rating the process as practical and acceptable. Attar et al. (2023) further demonstrated that electronic administration of the M-CHAT-R/F among 17,900 toddlers improved completion rates and eliminated the delay between the initial questionnaire and follow-up, although the authors emphasized that skilled clinical interpretation remains necessary.

Beyond individual tools, Huda et al. (2024) highlighted that while ASD screening instruments generally demonstrate strong validity and reliability, their effectiveness may vary across culturally and linguistically diverse populations. This underscores the need for contextual adaptation of screening

tools to ensure accuracy and inclusivity, particularly in low-resource settings.

The thirteen behavioral indicators adopted in the present study are likewise grounded in empirical evidence. Chang et al. (2021), using computational gaze analysis of 993 toddlers aged 17 to 37 months, documented reduced attention toward social stimuli and differences in the coordination of gaze with speech sounds among children with ASD, providing support for the eye contact indicator. Delehanty and Wetherby (2021), observing 211 children of approximately 20 months during everyday home activities, found that the ASD group used a smaller variety of gestures and showed lower rates of deictic gestures such as pointing, supporting the gesture use, joint attention, sharing interest, and language indicators. Wu et al. (2024) extended this finding longitudinally, reporting that among 467 children assessed at 12, 18, and 24 months, those later diagnosed with ASD showed slower growth in deictic and symbolic gestures. Lee and Schertz (2022) distinguished social from instrumental turn-taking among toddlers with autism and found that only social turn-taking was positively associated with joint attention, reinforcing the importance of assessing the social purpose of a child's communication rather than its frequency alone.

With respect to the restricted and repetitive behavior domain, Miller et al. (2021) found that more frequent unusual visual inspection of objects was observable in children later diagnosed with ASD as early as nine months, supporting the play behavior and repetitive behavior indicators. Chaxiong, Dimian, and Wolff (2022), in a systematic review of forty-one studies covering birth through age three, reported that differences in the frequency and intensity of repetitive behaviors are detectable before age two, providing the

broadest support for the repetitive behavior, sensory response, emotional response to change, and fixated interest indicators. The same review cautioned, however, that findings on certain sensory behaviors and insistence on sameness remain mixed, and that the quantity of repetitive behaviors should not be treated as a direct measure of autism severity.

In parallel, emerging technologies have shown strong potential in improving ASD detection. Studies by Saihi et al. (2021) demonstrated that machine learning can enhance prediction accuracy in behavioral classification. Rubio-Martín et al. (2024) showed that Natural Language Processing (NLP) can effectively analyze language patterns associated with autism, while Deng et al. (2024) introduced chatbot-based systems powered by large language models, improving accessibility and user interaction in screening processes. Collectively, these findings suggest that integrating artificial intelligence with traditional screening approaches can improve both efficiency and usability.

Several studies have applied supervised machine learning to questionnaire-based toddler screening data. Sollis, Wall, and Washington (2025) applied Decision Tree, Random Forest, and XGBoost models to the QCHAT-10 dataset and reported up to 91% AUC using fewer screening items. Tartarisco et al. (2021) compared five algorithms, including Random Forest and Logistic Regression, on Q-CHAT data, with Support Vector Machine achieving the highest accuracy at 95%. Lu et al. (2024) evaluated seven supervised models on the CHAT-23 questionnaire and found higher diagnostic accuracy among children under 36 months. These findings support the use of tree-based, boosting, and linear classifiers on parent-reported questionnaire

data. The present study limits its scope to Logistic Regression, Random Forest, and XGBoost combined through a Soft Voting Ensemble, which averages the predicted class probabilities of the three models so that the final classification reflects the confidence of each model.

However, despite these advancements, local literature in the Philippines remains focused primarily on the social, emotional, economic, and systemic challenges faced by individuals with ASD and their families. Studies by Balbin et al. (2022), Chiu et al. (2023), Reyes and Domingo (2024), and Dy et al. (2023) consistently highlight barriers such as limited access to healthcare services, financial constraints, and insufficient institutional support. Quilendrino et al. (2022) quantified this burden directly, reporting among 158 Filipino families an average diagnostic evaluation cost of approximately ₱7,411.80 and average first-year post-diagnosis intervention expenses of ₱38,868, with costs varying according to family income and access to services. Leosala (2023) similarly documented, through interviews with ten Filipino parents, that families actively seek information and adapt to differing support needs while facing persistent service barriers.While these studies emphasize the need for early detection, only a few local studies explore technology-based approaches. For instance, de Vera et al. (2024) examined intelligent systems for monitoring and predicting autism-related behaviors, but not a parent-facing, guided pre-screening system for toddlers.

Overall, the literature reveals a clear gap between validated ASD screening instruments, emerging AI-based detection technologies, and their integration into accessible local systems. Although international studies demonstrate the effectiveness of combining standardized screening tools with

machine learning, NLP, and chatbot systems, and although the individual behavioral indicators used in the present questionnaire are each supported by empirical research, such integration has not yet been fully implemented in the Philippine context. This gap highlights the need for a culturally appropriate, accessible, and technology-driven ASD pre-screening system that can support early identification, particularly among toddlers in underserved communities.

CHAPTER 3

METHODOLOGY

Methods of Research Used

This study employs an experimental research method that integrates quantitative and qualitative approaches to develop and evaluate the AuRA (Autism Risk Analytics) system for Autism Spectrum Disorder (ASD) pre-screening among toddlers aged 12 to 36 months.

The study uses an experimental research design because it focuses on developing, testing, comparing, and evaluating different supervised machine learning algorithms for classifying ASD risk levels. Specifically, the study evaluates Logistic Regression, Random Forest, and Extreme Gradient Boosting (XGBoost) based on parent-reported behavioral information. The predictions of these models are combined using a Soft Voting Ensemble to produce the final risk classification.

The quantitative approach is used to measure the performance of the machine learning models using numerical data and statistical measures. The structured responses from the questionnaire are assigned numerical scores, while the parent-reported open-ended responses are processed through Natural Language Processing (NLP) and Behavioral Indicator Matching. The identified behavioral indicators are converted into numerical feature vectors using Ordinal Feature Encoding, where each indicator is represented using values of 0, 1, or 2 corresponding to the defined level of behavioral observation or screening concern. The performance of the machine learning

models is evaluated using accuracy, precision, recall, F1-score, and confusion matrix.

The qualitative approach is used because the questionnaire includes open-ended follow-up questions that allow parents or legal guardians to describe their child's behaviors in their own words. These textual responses provide detailed behavioral information that serves as input for the NLP component of the AuRA system. The questionnaire is based on the Modified Checklist for Autism in Toddlers, Revised with Follow-Up (M-CHAT-R/F), Quantitative Checklist for Autism in Toddlers (Q-CHAT), and Communication and Symbolic Behavior Scales Developmental Profile Infant-Toddler Checklist (CSBS-DP ITC), with additional behavioral indicators incorporated based on related literature and expert recommendation.

Research Design

This study employs an experimental research design integrating quantitative and qualitative approaches to develop and evaluate the AuRA (Autism Risk Analytics) system for Autism Spectrum Disorder (ASD) pre-screening among toddlers aged 12 to 36 months.

The researchers developed a questionnaire based on three established ASD screening tools: the Modified Checklist for Autism in Toddlers, Revised with Follow-Up (M-CHAT-R/F), the Quantitative Checklist for Autism in Toddlers (Q-CHAT), and the Communication and Symbolic Behavior Scales Developmental Profile Infant-Toddler Checklist (CSBS-DP ITC). Through comparative analysis, related literature, and expert recommendation, **thirteen (13) ASD behavioral indicators** were established as the foundation of the

questionnaire and the system's ASD Domain Knowledge. The thirteenth indicator was incorporated based on the recommendation of a qualified medical expert during the review of the questionnaire.

The questionnaire consists of **main questions with structured response choices and open-ended follow-up questions**. The structured responses are used for numerical scoring, while the open-ended follow-up responses provide detailed behavioral information for the NLP component of the system.

The experimental component focuses on evaluating the classification performance of Logistic Regression, Random Forest, and XGBoost in classifying ASD risk levels based on the collected parent-reported behavioral information. Since no existing dataset fully matches the objectives of the study, the researchers collect and develop their own dataset from qualified parents or legal guardians.

Before machine learning classification, the collected open-ended responses undergo Natural Language Processing (NLP) through text preprocessing to organize and standardize the textual data. The processed responses then undergo Behavioral Indicator Matching using the system's ASD Domain Knowledge to identify and interpret the **thirteen (13) ASD behavioral indicators** and represent their observed levels using ordinal numerical values. The identified indicators are converted into numerical feature vectors using an Ordinal Feature Encoding. The dataset is then assigned ASD risk labels through Ground Truth Labeling for supervised machine learning.

The labeled dataset is used to train and test the Logistic Regression, Random Forest, and XGBoost models. The predictions generated by the three models are combined using a Soft Voting Ensemble to produce the final ASD risk classification. The performance of the individual models and the ensemble is evaluated using accuracy, precision, recall, F1-score, and confusion matrix.

The AuRA system also incorporates a chatbot interface as the main platform for user interaction and ASD pre-screening. The chatbot guides respondents through the questionnaire, collects their structured and open-ended responses, and processes the collected information through the NLP, Behavioral Indicator Matching, Ordinal Feature Encoding, and machine learning pipeline. The chatbot also provides evidence-based information about ASD through the system's Medical Library.

Locale of the Study

This study was conducted through online and paper-based data collection. The online questionnaire was administered using KoboToolbox, while printed copies of the researcher-developed questionnaire were provided to qualified participants who participated through paper-based data collection.

The data collection will not be limited to a specific institution or geographical location. Qualified parents or legal guardians of toddlers aged 12 to 36 months from different communities may participate, provided that they meet the established inclusion criteria.

The online approach will allow participants to answer the questionnaire using internet-enabled devices such as smartphones, tablets, laptops, or desktop computers. Meanwhile, the paper-based approach will provide an alternative for participants who may not have convenient access to an internet-enabled device or may prefer to answer a printed questionnaire.

Both the online and paper-based questionnaires will contain the same researcher-developed questionnaire, including the thirteen (13) ASD behavioral indicators, structured response choices, and open-ended follow-up questions. The collected responses from both methods will be consolidated into a single dataset for data processing, NLP, scoring, and machine learning analysis.

The use of both online and paper-based data collection is appropriate for the study because it allows the researchers to reach qualified participants through different methods while maintaining consistency in the research instrument and data collection process.

Respondents of the Study

The primary respondents of this study are parents or legal guardians of toddlers aged 12 to 36 months. They are selected because they have regular and direct interaction with their children and can provide observations regarding their child's communication, social interaction, play behavior, language development, sensory responses, repetitive behaviors, and other developmental characteristics related to ASD.

To participate in the study, respondents must meet the following inclusion criteria:

- Must be the biological parent or legal guardian of a child aged 12 to 36 months.
- Must have regular and direct interaction with the child.
- Must be able to understand and answer the researcher-developed questionnaire.
- Must voluntarily provide informed consent before participating.
- Must have access to an internet-enabled device to answer the online questionnaire, or must be able to answer the printed questionnaire.

Responses that do not satisfy the inclusion criteria or contain incomplete questionnaire data are excluded from the final dataset.

In addition to the parent respondents, the study involves one Developmental Pediatrician who serves as the expert validator. The Developmental Pediatrician reviews the researcher-developed questionnaire, the thirteen (13) ASD behavioral indicators, scoring system, and proposed risk criteria. The expert's feedback and recommendations are used to improve the clarity, relevance, appropriateness, and validity of the research instrument and related methodology.

Research Instruments

The primary research instrument used in this study is a researcher-developed ASD pre-screening questionnaire based on the

Modified Checklist for Autism in Toddlers, Revised with Follow-Up (M-CHAT-R/F), Quantitative Checklist for Autism in Toddlers (Q-CHAT), and Communication and Symbolic Behavior Scales Developmental Profile Infant-Toddler Checklist (CSBS-DP ITC). The questionnaire is written in English / Filipino, and respondents may answer the open-ended follow-up questions in English, Filipino, or Taglish. The system's behavioral lexicon includes English, Taglish, and Tagalog expressions to account for this.

Through comparative analysis of the established screening tools, related literature, and expert recommendation, thirteen (13) ASD behavioral indicators were identified and incorporated into the questionnaire.

The questionnaire consists of thirteen main questions with structured response choices and open-ended follow-up questions. The main questions are used for numerical scoring, while the open-ended follow-up questions are designed to collect detailed descriptions of the child's behavior. These textual responses serve as data for the NLP component of the AuRA system.

The questionnaire underwent pilot testing to identify questions that were difficult for parents to understand, inconsistencies in response choices, and other issues in answering the questionnaire. Based on the pilot findings and expert recommendations, revisions were made to improve the clarity and appropriateness of the questionnaire before its use for the main data collection.

The questionnaire also uses a scoring system in which Q1–Q9 use Always = 0, Sometimes = 1, and Never = 2, while Q10–Q13 use Yes = 2, Sometimes = 1, and No = 0, which is a reversal of the ordinality of the former.

The resulting scores are used as structured behavioral information and are considered together with the processed textual responses.

Data Gathering Procedure

The researchers first obtained the necessary approval from the thesis adviser, panel members, and appropriate school authorities before conducting the data gathering.

The researcher-developed questionnaire was developed based on the Modified Checklist for Autism in Toddlers, Revised with Follow-Up (M-CHAT-R/F), Quantitative Checklist for Autism in Toddlers (Q-CHAT), Communication and Symbolic Behavior Scales Developmental Profile Infant-Toddler Checklist (CSBS-DP ITC), related literature, and the recommendation of a qualified Developmental Pediatrician. The questionnaire was reviewed by the Developmental Pediatrician to assess the clarity, relevance, appropriateness, and completeness of the items.

A pilot test was conducted to determine whether the parents could understand the questions and response choices and to identify items that required clarification or revision. Based on the pilot findings and expert recommendations, revisions were made to improve the questionnaire before the main data collection.

For the main data gathering, the researchers used two methods: online and paper-based data collection. The online questionnaire was administered through KoboToolbox, while printed copies of the same questionnaire were provided to participants who answered through the paper-based method.

Before answering the questionnaire, respondents were provided with an informed consent form. Only participants who voluntarily agreed to participate and met the established inclusion criteria were allowed to proceed with the questionnaire.

The questionnaire consisted of thirteen (13) main questions with structured response choices and open-ended follow-up questions. The structured responses were collected for scoring, while the open-ended responses were collected to provide detailed descriptions of the children's behaviors for the NLP component of the proposed AuRA system.

After data collection, the responses from the KoboToolbox and paper-based questionnaires were consolidated into a dataset. The researchers then reviewed the collected responses for completeness and consistency and organized the data for subsequent preprocessing, scoring, NLP processing, behavioral indicator matching, and machine learning analysis.

Proof of Concept

The Proof of Concept (PoC) demonstrates the technical feasibility of AuRA as an Autism Spectrum Disorder (ASD) pre-screening system. It illustrates how parent-reported behavioral observations are processed through Natural Language Processing (NLP), transformed into numerical feature representations, and prepared for machine learning classification. The chatbot serves as a guided assessment interface and provides educational information about ASD through the system's Medical Library.

The proposed workflow follows an NLP–ML–NLP structure, consisting of the following stages:

1. Input and Initialization – The system collects the necessary respondent and child information and presents the structured assessment questions. Parents or legal guardians provide structured answers and open-ended descriptions of their children's observed behaviors.

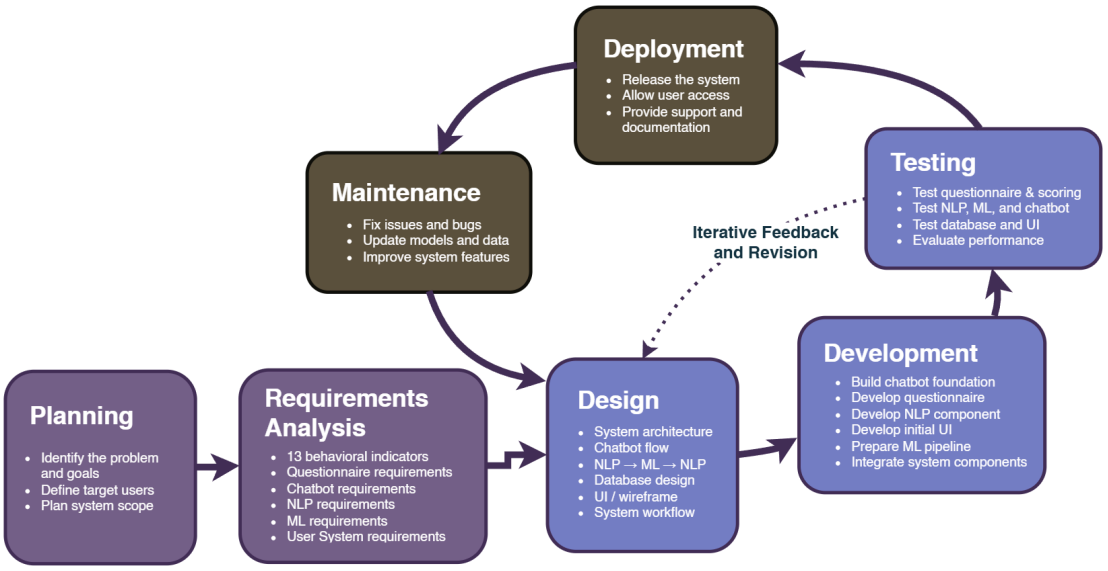
2. NLP Processing and Feature Extraction – The system normalizes and tokenizes open-ended responses, then applies rule-based behavioral indicator matching using the ASD Domain Knowledge Base. Context, polarity, and frequency interpretation are used to determine the appropriate representation of each identified behavioral indicator. The resulting observations are converted into ordinal numerical values of 0, 1, or 2 and organized into a Numerical Feature Vector (NFV).

3. Machine Learning Classification – The resulting NFV serves as the input to the machine learning component. Logistic Regression, Random Forest, and XGBoost are intended to generate class predictions or probabilities based on their trained models. Under the selected Soft Voting Ensemble approach, the predicted class probabilities are averaged, and the class with the highest average probability determines the final classification. This stage depends on the availability of a properly labeled dataset and trained models.

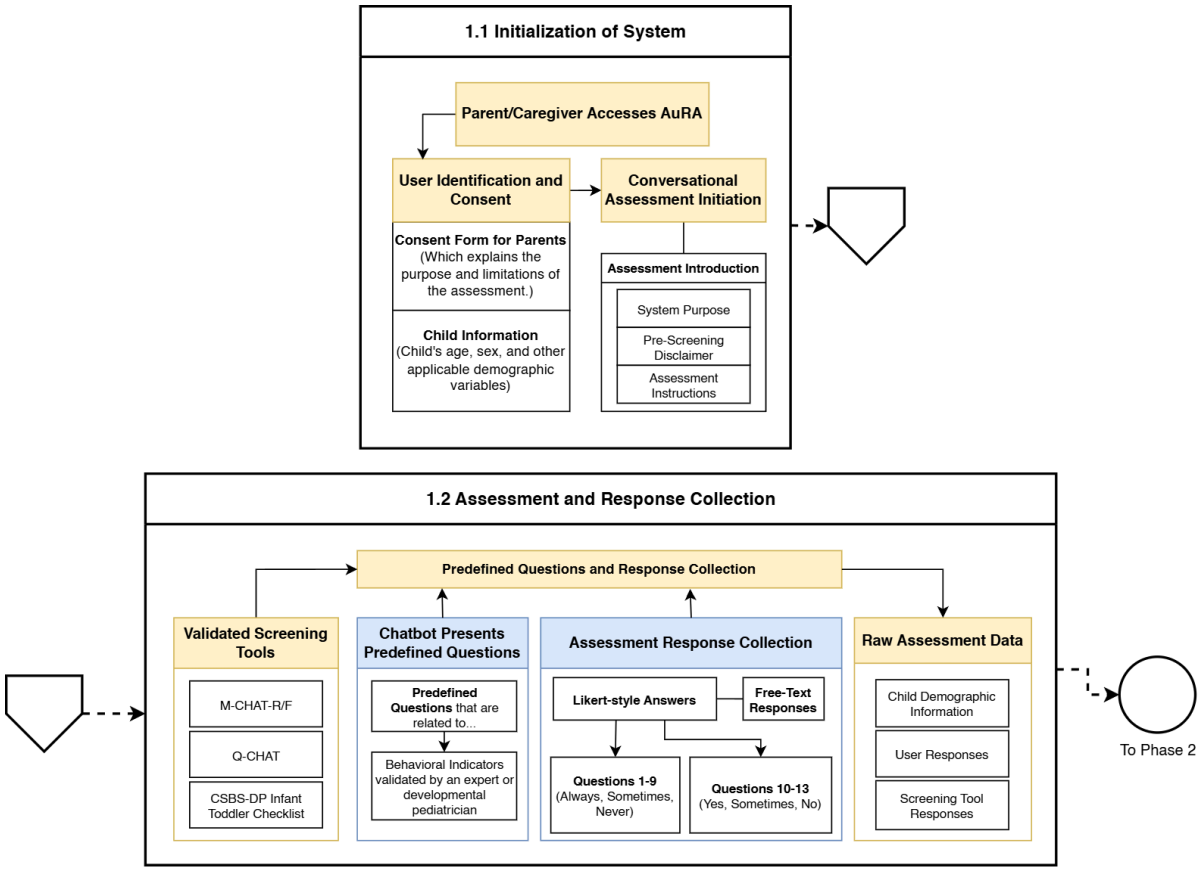
4. Response Generation and Educational Feedback – The system presents the assessment outcome through predefined, reviewed response templates. It may also provide relevant ASD educational information through the Medical Library and guidance to seek professional evaluation when appropriate. AuRA is intended for pre-screening and awareness support and does not provide a clinical diagnosis.

The PoC is evaluated by examining whether the system can receive sample responses, process behavioral descriptions, produce the expected ordinal feature representation, and pass valid feature vectors to the classification component when the trained models are available. The functionality demonstrated in the prototype is distinguished from components that remain under development.

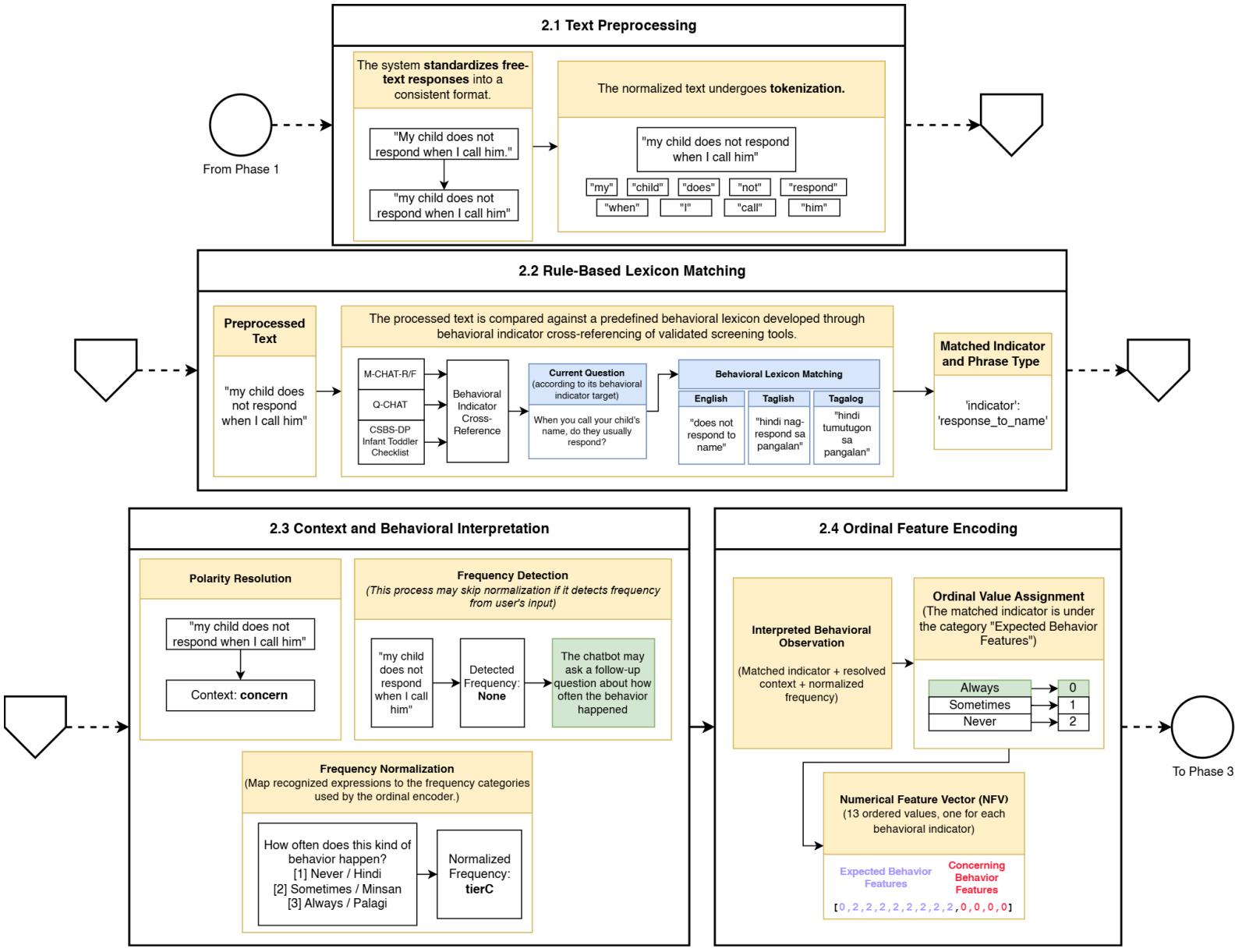
Software Model



Phase 1: Input and Initialization



Phase 2: Data Processing and NLP Feature Extraction



Phase 3: Machine Learning Layer

